

How Challenges Become Opportunities: Micro-credentials and Artificial Intelligence

Carlos Delgado Kloos
Dept. Telematic Engineering
Universidad Carlos III de Madrid
E-28911 Leganés (Madrid), Spain
ORCID: 0000-0003-4093-3705

Carlos Alario-Hoyos
Dept. Telematic Engineering
Universidad Carlos III de Madrid
E-28911 Leganés (Madrid), Spain
ORCID: 0000-0002-3082-0814

Rebiha Kemcha
Dept. Telematic Engineering
Universidad Carlos III de Madrid
E-28911 Leganés (Madrid), Spain
ORCID: 0000-0002-9540-7887

Pedro Manuel Moreno-Marcos
Dept. Telematic Engineering
Universidad Carlos III de Madrid
E-28911 Leganés (Madrid), Spain
ORCID: 0000-0003-0835-1414

Iria Estévez-Ayres
Dept. Telematic Engineering
Universidad Carlos III de Madrid
E-28911 Leganés (Madrid), Spain
ORCID: 0000-0002-1047-5398

Patricia Callejo-Pinardo
Dept. Telematic Engineering
Universidad Carlos III de Madrid
E-28911 Leganés (Madrid), Spain
ORCID: 0000-0001-6124-6213

Pedro J. Muñoz-Merino
Dept. Telematic Engineering
Universidad Carlos III de Madrid
E-28911 Leganés (Madrid), Spain
ORCID: 0000-0002-2552-4674

María-Blanca Ibáñez
Dept. Telematic Engineering
Universidad Carlos III de Madrid
E-28911 Leganés (Madrid), Spain
ORCID: 0000-0003-2612-4207

Mario Muñoz-Organero
Dept. Telematic Engineering
Universidad Carlos III de Madrid
E-28911 Leganés (Madrid), Spain
ORCID: 0000-0003-4199-2002

Abstract—As we move from the Information Age to the Intelligence Age, universities must redefine themselves taking into account recent challenges. Two of these challenges are Micro-credentials and Artificial Intelligence (AI). Micro-credentials certify the learning outcomes of short-term learning experiences, which are typically more flexible and tailored to specific, job-relevant skills, meeting the increasing demand for continuous education. These learning experiences are normally referred to as micro-credential courses or micro-credential programs and disrupt traditional degree models. On the other hand, Artificial Intelligence is disrupting every single aspect of our life and work. Artificial Intelligence tools can work as assistants helping in all kinds of tasks that were previously reserved for humans. Artificial Intelligence can be used in education in many ways, such as to create personalized learning paths, create and optimize educational multimedia content such as text, voice, image, or video, tutor students, and monitor student progress in real-time, contributing to the acquisition of learning outcomes.

Putting the two challenges together, Artificial Intelligence can also contribute to generate micro-credentials by ensuring learners demonstrate competence in practical, job-specific skills,

enhancing their credibility for employers. Therefore, Artificial Intelligence can play a pivotal role in advancing micro-credential courses and programs, helping universities redefine their offerings to provide personalized, adaptable, and industry-relevant learning experiences. Artificial Intelligence can help in designing the content and the learning experience as with courses in general, but it can also help in designing the specificities of micro-credential courses. The specific “skills” provided by Artificial Intelligence for defining micro-credentials range from clerical work related to mastering formats such as ELM or OpenBadges that are needed to construct digital credentials to creative suggestions that aid in designing the details of the micro-credential course. Alternatively, the micro-credentials designed can include topics about Artificial Intelligence, helping the workforce on this important area.

This paper analyzes the interplay of these two challenges and how they can help each other, making opportunities out of challenges.

Index Terms—Artificial Intelligence, Micro-credentials, Large Language Models, Learning Outcomes, Personalized Learning, Digital Credentials.

I. INTRODUCTION

Micro-credential courses are short, focused courses intended to reskill and upskill the workforce. The European Commission says that 60% of the workforce should receive such courses every year [1]. UNESCO [2] and the OECD [3] reinforce the importance of this need. Micro-credentials certify the learning outcomes obtained by learners after completing such micro-credential courses or programs (depending on the granularity and learners’ workload) [1]. As Prof. Mauro Calise (Emeritus Professor at University of Naples Federico II, Italy)

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says, the university has to redefine itself from mainly serving students for 3+2 years to serving additionally workers for 30+20 years. This is a major challenge, since although lifelong learning and professional development courses have always been offered, the emphasis is now different and, therefore, a major internal reorganization is needed [4]. Micro-credential courses have a shorter life, they should be launched and removed at high speed, or updated as the context changes. They are typically more multidisciplinary than traditional courses. The typical department-based organization with long approval times and preparation of material does not fit so well [5]. Agile procedures are needed to face this challenge.

Another big challenge for universities is how to deal with Artificial Intelligence (AI). Although Artificial Intelligence is an old concept [6], it has arrived in everyone's lives when OpenAI released the ChatGPT chatbot on 30 November 2022 [7]. After a first moment of denial and prohibition in universities due to its possible fraudulent use, it became clear very soon that students should be trained on how to take advantage of its benefits, since in their professional work they would need to use it [8]. Following the release of ChatGPT, there has been a rapid evolution of AI models (and particularly Generative AI) that support a variety of input and output formats, including text, image, audio, and video. In addition, numerous tools have emerged that extend the capabilities of these models by adding enhanced functionality [9]. These tools can help educators, for example, design learning objectives and create materials, automate routine tasks, and monitor students' progress, and can also help students, for example, receive more personalized support and feedback. For both teachers and students, the effective use of these AI-powered tools is essential to fully unlock their educational potential and enhance learning experiences.

This paper explores the interplay between micro-credentials and AI, focusing on how AI can support the design and enactment of micro-credential courses and programs. Specifically, Section 2 defines a micro-credential workflow, which includes the design and enactment phases in a micro-credential course. Section 3 focuses on the various formats available for generating the actual digital credential (that is, the record of the learning outcomes), with particular emphasis on ELM (European Learning Model) [10]. Section 4 presents a case study illustrating the integration of Generative AI within the micro-credential workflow including the information needed to generate the credential itself following ELM. Finally, Section 5 presents the conclusion derived from this work.

II. MICRO-CREDENTIAL WORKFLOW

Figure 1 shows the complete workflow for a micro-credential course from its design through its enactment till the final certification of student learning outcomes, based on the performance, with a digital credential. In the design phase, the teacher defines the learning objectives and creates educational materials aligned with these objectives. During the enactment phase, the teacher delivers the content, using the appropriate format (whether synchronous or asynchronous, face-to-face,

online, or hybrid) and technologies (tools, platforms, etc.) to support effective teaching. Meanwhile, the student engages in the course, participates in classes, demonstrates mastery of the subject matter through performance, and ultimately gets the micro-credential that certifies the learning outcomes. The micro-credential is typically issued in a digital format, enhancing both its portability and verifiability.

This workflow may apply to many types of courses, but in the case of micro-credential courses it should have a fast turn-around. The implementation of this workflow demands a more agile approach compared to traditional, longer academic programs. We want to analyze now how AI can help precisely to implement an agile approach in each of the different phases of this micro-credential workflow.

It has already been discussed how AI can help teachers in the design and enactment in class, as well as how AI can assist learners in their learning [11]. In particular, the use of Generative AI for teachers was classified into three main purposes: 1) understanding the topics to teach (research for course preparation, summaries of documents and videos, and proposals for the syllabus); 2) preparation of educational material (documents, mindmaps, slides, images, videos, and assessments and rubrics); and 3) enactment (teaching methods, orchestration of class, feedback to learners, and real-time support to instructors). The use of Generative AI for learners in the enactment phase was also classified on several purposes: Rephrasing and interpretation (of text, images, and other types of learning materials), getting personalized content, mentoring, support as teaching assistant in online forums, support to special education students, role-playing as an additional learner to whom to explain a topic (peer teaching), support to self-reflection of metacognitive skills, and detection of students' emotions. In addition, there is already a lot of reflection on how to rethink assessment in educational settings and how to measure student performance in times of Generative AI [12].

The large box with a green background corresponds to the learning opportunity and has been traditionally kept hidden, encapsulated. It includes the learning material, the enactment and the work of the student, whereas the two other boxes correspond to the interface to the outside. They are the high-level specification of the course goals, particularly the planning of the learning outcomes (first box in the workflow at the top left) and the certification of the student performance (i.e., the certification of the achieved learning outcomes) with a digital credential (last box in the workflow in the bottom left). For these two boxes, it is important that they are public, whereas the four other boxes (with the green background) are typically private. Exceptions to this are Open Educational Resources (OER), where the educational material is also made public and open lectures where the enactment is also public.

The two boxes in a large box with a dark blue background are also related. They can be related to specification and design in software engineering. The "Planned Learning Objectives" correspond to the specification and the "Creation of the Educational Material" to the design of a software program. Enactment corresponds to the execution of the program. These

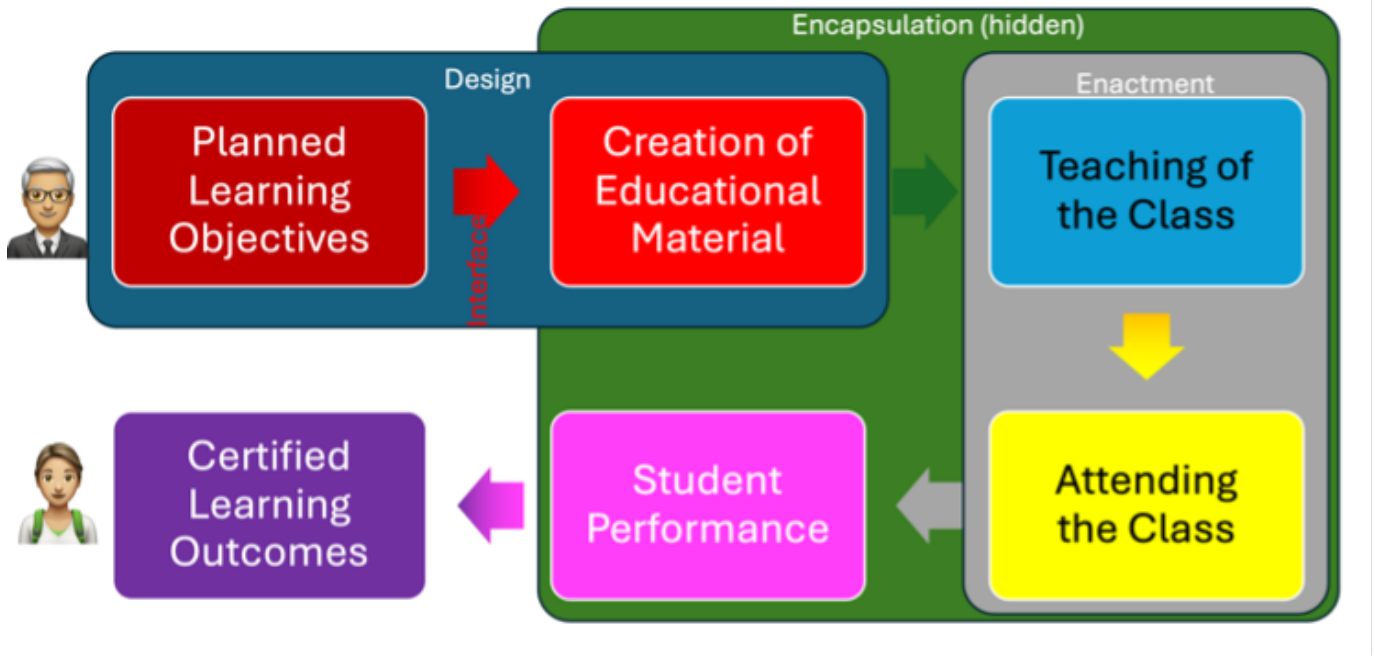


Fig. 1: Workflow from the design and enactment of a micro-credential course by the teacher to the certificate obtained by the student with the learning outcomes of this course.

parallels are interesting, because we can apply software engineering techniques to course development. Especially with micro-credential courses, reusability and evolution of course designs is important due to the fast pace of change. We will not analyze this venue in this paper. What we will analyze is how AI can help us cover the two boxes on the left for objectives and outcomes. To do this, we must first analyze the existing formats for issuing digital credentials and how they affect the definition of learning objectives and outcomes in micro-credential courses.

III. FORMATS FOR DIGITAL CREDENTIALS

The successful passing of a course by a student could typically be certified with a paper diploma. Diplomas and certificates on paper have problems with fraud [13]. Nowadays, certificates can be issued in digital form [14]. They can be printed to create a paper-based version, but more importantly, they can be easily verified, shared, and stored digitally. Moreover, they do not have the physical limitations of paper-based diplomas, so much more information can be stored in the digital counterpart.

There are several formats for digital credentials [15]. Two of the more prominent are OpenBadges [16] and ELM (European Learning Model) [10]. OpenBadges 3.0 and ELM v3 are based on W3C VC (Verifiable Credentials) [17], so there are some similarities between them [18]. The interoperability between these formats is being addressed. We will focus on ELM here.

ELM was designed to capture all aspects of the learning process with the aim to facilitate data interchange across

Europe. This is achieved through a comprehensive ontology [19]. Particularly important in this ontology are these concepts:

- Learning Achievement (acquisition of knowledge and/or skills),
- Learning Activity (action which may lead to the acquisition of knowledge and/or skills),
- Learning Assessment (the process of establishing the extent to which a learner has attained particular knowledge, skills and competences),
- Learning Entitlement (a right, e.g., to practice a profession or join an organization, as a result of the acquisition of knowledge and/or skills),
- Learning Opportunity (an opportunity to realize a given set of learning outcomes via a learning activity and/or assessment) and
- Learning Outcome (a statement regarding what a learner knows, understands and is able to do, defined in terms of knowledge and/or skills).

Therefore, it is noteworthy that the learning outcomes are a key part of the information contained in the certificate that the learner obtains upon completion of the micro-credential course.

It is also very important to distinguish between learning objectives and learning outcomes, as the latter are the ones contained in a digital certificate that complies with ELM. *Learning objectives* are specific goals that outline what the instructor intends to teach or what learners are expected to understand during a lesson, course, or program. These objectives are instructor-centered, representing the intended goals for a learning experience. In contrast, *learning outcomes*

describe what the learners have actually accomplished at the end of a learning activity, course, or program. They are learner-centered, focusing on demonstrated knowledge, skills, or competencies. Learning outcomes are typically evaluated, with evidence collected to confirm that the learner has met the intended outcome.

IV. CASE STUDY

In this section, we present a case study of the design of a micro-credential course using AI. The topic of the course is an introduction to artificial intelligence. The AI tools we used for this experiment are ChatGPT with OpenAI o3-mini, ChatGPT with GPT-4 Turbo [20], DeepSeek-R1 [21], and NotebookLM [22]. The o3-mini is a small reasoning model from OpenAI, providing high intelligence at low cost and low latency, while excelling at scientific, mathematical and coding tasks. At the time of writing, this model is not free. As for the GPT-4 Turbo, this is a version of the GPT-4 pre-trained generative transformer with faster response times and high accuracy for rapid interaction. This AI model is available in the free version of ChatGPT.

DeepSeek-R1 is a cost-effective, open-source reasoning model with high-performance logic and chatbot functionality by DeepSeek. Finally, NotebookLM is an AI tool powered by Google and uses the Gemini 2.0 AI model as its Large Language Model (LLM).

The Guidelines for the Design and Accreditation of University Micro-credentials [23] and *The Guidelines for the Issuance of Digital Credentials for University Micro-credentials* [24] are taken as a starting point to work with the AI tools. In particular, the elements for the design and certification of a university micro-credential are considered, as presented in Table I. These elements are divided into three large blocks, for a total of 24 elements: 1) Identification (5 elements), 2) Description of the Learning Experience (12 elements), and 3) Assessment and Certification (7 elements). These elements are based on those presented in the European approach to micro-credentials [25].

It is important to highlight here, for example, the planning of the *learning objectives* as part of the design of the micro-credential course, as indicated in the Micro-credential Workflow in Figure 1. Expected learning outcomes must also be defined, and related, as possible, with existing models or frameworks, such as ESCO (European Skills, Competences, Qualifications and Occupations) [29], or DIGCOMP (European Digital Competence Framework for Citizens) [30].

A. Approach for the definition of the micro-credential course

In the approach, the files with the *Guidelines for the Design and Accreditation of University Micro-credentials* [23], the *Guidelines for the Issuance of Digital Credentials of University Micro-credentials* [24], and the file *Towards a Common Definition of Micro-credentials* [2] were uploaded into the AI tools. The tools were then asked to design a university micro-credential on an introduction to Artificial Intelligence, issued by Universidad Carlos III de Madrid. Additionally, a specific

task was requested from the AI tools namely the definition of the learning outcomes precisely using ESCO (see Listing 1). This is a complex task due to the large number of Skills, Competences and Occupations present in this framework and can be very time consuming for the designers of the micro-credential. The AI tools used the reference table from the Guidelines for the Design and Accreditation of University Micro-credentials and populated the 24 fields.

The entire prompt used in this experiment is as follows:

```
Create a university-level micro-credential program on "
Artificial Intelligence", following the guidelines
outlined in the documents "Orientaciones para el
diseño y acreditación de las Microcredenciales
Universitarias" and "Orientaciones para la Emisión
de Credenciales Digitales para las Microcredenciales
Universitarias".

Ensure that the micro-credential meets European quality
standards and includes all essential elements. The
program must comply with the principles set forth in
the uploaded document, including employability
focus, quality assurance, and modularity. It should
be designed for lifelong learners and professionals
seeking upskilling opportunities.

The micro-credential is intended for beginner level
learners and it will be issued by the university
UC3M.

Use ESCO in defining the learning outcomes (https://esco.
ec.europa.eu/en/classification/skill\_main).

Refer to the micro-credential example (page 16/32) in "
Orientaciones para la Emisión de Credenciales
Digitales para las Microcredenciales Universitarias"
to understand the required elements and structure.

Use angle brackets (< >) to indicate placeholders for
information that needs to be filled in later.

Structure of the micro-credential design using the
following elements and descriptions:
1- Identification
  - Identification of Recipient
  - Supervision and Verification of Identity
  - Name of the Micro-credential
  - Issuing University
  - Entity for Learning Experience
2- Description of the Learning Experience
  - Start and End Date
  - Language
  - Admission Requirements
  - Learning Objectives
  - Content Description
  - Mode of Instruction
  - Learning Outcomes
  - Level of Experience
  - Competency Frameworks
  - Learning Activities
  - Workload (Theoretical)
  - Workload (Practical)
3- Assessment and Certification
  - Assessment Methods
  - List of Trainers
  - Stackability Options
  - Outcome or Rights for the Student
  - Quality Assurance
  - Certifying University
  - Date of Issue
```

Listing 1: Prompt used for micro-credential definition

Tables II, III, and IV show the result of the design of the micro-credential, created with the support of the four AI tools used and considering these 24 elements. This example serves

TABLE I: Elements and Descriptions for the Design and Certification of a University Micro-credential (translated from [23]) divided in three categories: 1) Identification, 2) Description of the Learning Experience, 3) Assessment and Certification

1. Identification	
Identification of Recipient	Name of the individual receiving the certification.
Supervision and Verification of Identity	Type of identity verification during the course and evaluation (e.g., unsupervised, supervised online/in-person with identity verification).
Name of the Micro-credential	Title or denomination of the university micro-credential.
Issuing University	The university or universities offering the micro-credential.
Entity for Learning Experience	If different from the issuing university, name of the organization or institution where the learning occurs.
2. Description of the Learning Experience	
Start and End Date	The beginning and completion dates of the educational activity.
Language	Language of instruction.
Admission Requirements	Prerequisites for access and criteria for admission (e.g., level of prior education or knowledge).
Learning Objectives	Clearly defined objectives based on the target audience and the purpose of the training.
Content Description	Detailed description of the educational content.
Mode of instruction	In-person, online, or hybrid.
Learning Outcomes	Expected learning outcomes and type of achievement (e.g., skills, knowledge, competencies).
Level of Experience	Qualification level as per frameworks like EQF [26], and the Spanish MECU [27] or MECES [28].
Competency Frameworks	Competency frameworks used (e.g., ESCO [29], DIGCOMP [30], or others).
Learning Activities	Types of learning activities (theoretical and practical).
Workload (Theoretical)	Volume of theoretical work required to achieve the learning outcomes (in ECTS and hours, if applicable).
Workload (Practical)	Volume of practical work required to achieve the learning outcomes (in ECTS and hours, if applicable).
3. Assessment and Certification	
Assessment Methods	Type of assessments validating learning outcomes (e.g., skill application, video presentation, portfolio).
List of Trainers	Qualifications and professional experience of trainers.
Stackability Options	Possibility of integration with other micro-credentials (independent, integrated, stackable).
Outcome or Rights for the Student	Resulting credential, badge, program admission, or qualification earned.
Quality Assurance	Type of quality assurance or validation procedure used.
Certifying University	The legal name, mandatory legal identifier, and country/region of the university issuing the certification.
Date of Issue	Date on which the micro-credential is issued.

as a first draft of the design of the micro-credential course using an AI tool.

An analysis of the designs of the micro-credential course reveals that the 24 expected elements have been properly generated across the different models. Elements that need to be filled in further are beside angle brackets for easy identification, such as: the name of the student receiving the micro-credential (Identification of Recipient), the list of instructors responsible for its design and implementation (List of Trainers), the start date (Start Date) and end date (End Date), and the issuance date (Date of Issue), among others.

The four micro-credential definitions present distinct approaches in terms of content, structure and focus. The o3-mini model attempts to link the learning outcomes mapped by the ESCO and to respect its classification (skills, knowledge, etc.), but remains imprecise. This model favors ease of access for new learners, as well as the modularity of the micro-credential. It also offers flexibility in delivery modes—hybrid, virtual or face-to-face.

DeepSeek-R1 defines the micro-credential in a systematic way and draws a clear distinction between theoretical (2 ECTS) and practical (3 ECTS) workloads. It specifies detailed teaching modes with precise weightings (20% face-to-face, 80% virtual) and integrates blockchain-based digital authentication

to reinforce security and trust.

The GPT-4 Turbo model offers a concise yet structured approach, focusing on the application of AI in industry and supporting language adaptability in both Spanish and English. It offers a well-defined assessment plan that includes a theory exam, practical assignments and a peer review, each aimed at achieving Europass-compliant digital qualifications.

The NotebookLM adopts a more policy-oriented and theory-driven approach. It focuses on the definition of micro-credentials in the context of the files provided, which describe the UNESCO and European Union guidelines. While it is less prescriptive in terms of the nature of learning activities and recognition processes, it offers a more detailed presentation on quality assurance, course content and learning outcomes associated with learning experiences.

The four suggested definitions of quality assurance (see Table IV) vary in scope, specificity, and alignment with regulatory European standards, as mentioned in the used files. DeepSeek-R1 is more thorough than the other AI tools in recommending an “Annual review by the Teaching Innovation Committee” alongside “Audits by ANECA (National Agency for Quality Assessment and Accreditation).” NotebookLM adheres to ESG [32], while GPT-4 Turbo refers to EDC (European Digital Credentials) [33], but without containing

clear review procedures. ChatGPT with o3-mini model mentions only the Internal Quality Assurance System of UC3M SGIC(Sistema de Garantía Interna de Calidad) and general “European regulations.”

We find that the reasoning models outperform the GPT or LLM models, as shown by the specifications provided by DeepSeek-R1 and ChatGPT with o3-mini. Their provided designs are more detailed than those provided by NotebookLM and GPT-4 Turbo, although NotebookLM has followed the guidelines provided, producing results close to those of the reasoning models.

DeepSeek-R1 and NotebookLM provide a similar distribution of workload with the same number of ECT credits (5), taking 25 hours per credit as a reference. This number of credits is the one suggested in the example presented by the *Guidelines for the Issuance of Digital Credentials for University Micro-credentials* [24]. GPT-4 Turbo suggests fewer hours, which may be explained by the fact that document analysis was not done in depth, or that more iterations and/or context are required. ChatGPT with o3-mini, on the other hand, considers this information as a variable to be filled in.

While these initial drafts need to be refined and tailored to the specific needs of the instructors managing the micro-credential, they provide a good starting point. Key elements, such as the definition of learning objectives using action verbs, the organization of the syllabus into modules, the balance between theory and practice (and the workload of each part), and the assessment methods based on a portfolio of evidence and the presentation and defense of the micro-credential design project, are well-established. Specifically, the definition of learning outcomes and how these are assessed (see Assessment and Certification) are key to comply with the Micro-credential Workflow in Figure 1.

Building on this initial design, each module can be further developed using AI tools to support the micro-credential course creation process. These AI tools can, for example, assist in outlining detailed content by generating topic ideas, suggesting relevant readings, and helping to structure lessons according to the learning objectives. AI tools can also support the creation of educational materials, such as multimedia resources (videos, podcasts, etc.) quizzes, and assignments. AI tools can also offer insights into how to optimize these materials for different types of learners, providing also recommendations for integrating practical activities or interactive elements to find the right balance between theory and practice.

B. About the AI tools and Models

In this case, ChatGPT [20], DeepSeek [21], and NotebookLM [22] were used as the AI tools to support the design of a university micro-credential. Regardless of the specific AI tool employed, providing examples or contextual information is essential for achieving better results. Clear context enables the AI tool to generate more accurate and relevant outputs as part of the design process.

Although commonly known as “ChatGPT o3-mini” and “ChatGPT-4 Turbo,” their official designations are “OpenAI

3o-mini” and “GPT-4 Turbo.” Nonetheless, the ChatGPT platform is the primary interface where users engage with all the models provided by OpenAI.

The ChatGPT or OpenAI o3-mini is a reasoning model that uses a large language model trained using reinforcement learning to perform complex reasoning tasks. Thanks to this specialized training, the model outperforms conventional language models in terms of structured thinking and analytical problem-solving capabilities.

GPT-4 Turbo, released in December 2023, is an advanced GPT model optimized for chat interactions and text generation. Its training on extended datasets enables it to process multimedia inputs and natural language instructions, allowing it to generate a variety of results, including code, mathematical equations or human-like responses. This model is available free of charge via the ChatGPT chat interface.

DeepSeek-R1 is an open-source AI model specially designed for advanced reasoning tasks. Unlike traditional models, it skips the initial supervised learning phase and relies on reinforcement learning for training. Its training process comprises several stages, starting with carefully selected data before moving on to reinforcement learning. As a “Mixture of Experts” (MoE) model, DeepSeek-R1 is computationally efficient, activating only a subset of its system for each task, thus optimizing resource utilization.

The fact that DeepSeek-R1 is a serious competitor to other high-performance AI models, while being free and open-source, has made it a prime target for intensive use and tuning. This has led to numerous “out of order” situations, making its use uncertain.

DeepSeek-R1 and OpenAI o3-mini offer insight into their reasoning processes when generating responses. This transparency is extremely valuable for understanding “why” specific information is suggested. In our case, this allows us to refine the prompts and the resources used to guide the models.

Some aspects of DeepSeek-R1’s reasoning process are presented below.

```
Okay, I need to create a university-level micro-
credential program on "Artificial Intelligence"
following the guidelines from the provided Spanish
documents. Let me start by understanding what those
documents require.

First, the "Orientaciones para el diseño y acreditación
de las Microcredenciales Universitarias" emphasizes
employability, quality assurance, modularity, and
alignment with European standards. The micro-
credential should be part of lifelong learning,
targeting professionals seeking upskilling. Since it
's beginner-level, the content should be
foundational.

...

Next, the micro-credential must include details like
start/end dates, language (Spanish, probably),
admission requirements (maybe no prior degree needed
but basic math skills?), mode of instruction (
hybrid, as per the example), workload in ECTS (since
it's beginner, maybe 5 ECTS, which is 125 hours
total).

...
```

Listing 2: DeepSeek-R1’s reasoning process during micro-credential definition

TABLE II: Overview of “Identification” Definitions

Element	ChatGPT (o3-mini)	DeepSeek-R1	ChatGPT (GPT-4 Turbo)	NotebookLM (Gemini 2.0)
Recipient Identification	<To be verified through university enrollment system>	<Student’s full name>	<Recipient’s Full Name>	<Student’s Full Name>
Supervision and Verification	<To be determined by UC3M digital credentials policy>	Verification method (e.g., blockchain-secured ID checks during enrollment and assessments)	Digital ID Verification Platform/Procedure	Online supervision with identity verification, in accordance with quality assurance systems
Micro-credential Name	Microcredencial Universitaria en Inteligencia Artificial	University Micro-credential in Artificial Intelligence Fundamentals	Foundations of Artificial Intelligence for Beginners	University Micro-credential in Artificial Intelligence for Beginners
Issuing University	Universidad Carlos III de Madrid	Legal name: Universidad Carlos III de Madrid	Universidad Carlos III de Madrid (UC3M)	Universidad Carlos III de Madrid (UC3M)
Entity for Learning Experience	UC3M, Faculty of Engineering/Computing/AI	UC3M School of Engineering (collaboration with <Collaborating entity name, e.g., Company X>)	UC3M, in collaboration with <Industry Partner, if applicable>	UC3M, in collaboration with <Industrial/Academic Partner (if applicable)>

As we can see in Listing 2, the model sometimes infers elements that could be explicitly defined to avoid ambiguity. For example, the language of the micro-credential and the recommended allocation of credits could be specified in the prompt.

Finally, NotebookLM allows the creation of notebooks, which are independent projects where users can upload a series of documents and subsequently ask questions about them. One of the key advantages of NotebookLM is its ability to manage and analyze multiple sources of information within a single project (notebook).

NotebookLM uses the Gemini 2.0 AI model [34], developed by Google DeepMind and featuring agentic AI. This multi-modal model can process and understand text, images, audio, and video. Released in December 2024, Gemini 2.0 improved on its predecessor by enhancing reasoning, efficiency and factual accuracy.

The feature of analyzing sources is especially useful in the design of micro-credentials, where uploading examples can lead to more accurate and relevant design, drawing from past experiences and best practices. This feature also allows for a more iterative process, where instructors can continually refine the design of each of the elements in Tables II, III, and IV in case it is needed.

In this case, the resulting tables contain text that is somewhat more precise and adapted to the context of the course to be taught.

V. CONCLUSION

AI and micro-credentials represent two of the most significant challenges universities are faced with looking forward. As AI continues to evolve, its applications within higher education are increasing fast, influencing research activities, administrative processes, but also teaching and learning. Here we have shown how to harness AI for the design of micro-credentials.

Different AI tools—ChatGPT o3-mini, DeepSeek-R1, ChatGPT-4 Turbo, and NotebookLM—have been used to provide context and examples so that better results can be

obtained when getting a first draft of the micro-credential design. This is of course an iterative design process that should involve the instructors in order to refine this draft design taking into account the elements necessary to design a micro-credential and the existing formats for digital credentials.

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TABLE III: Overview of “Description of the Learning Experience” Definitions

Element	ChatGPT (o3-mini)	DeepSeek-R1	ChatGPT (GPT-4 Turbo)	NotebookLM (Gemini 2.0)
Start and End Dates	<To be determined>	<Start Date> to <End Date> (6 weeks)	<Start date> to <Completion date>	<Start Date> - <End Date>
Language	Spanish	Spanish (materials available in English)	English	English
Admission Requirements	No prerequisites or basic computer skills	No prior AI knowledge; basic math and digital literacy understanding	No prior knowledge of AI required; open to students and professionals	No prior qualifications required; interest in AI
Learning Objectives	- Introduce the fundamental concepts of artificial intelligence and machine learning. - Develop practical skills for the implementation of basic algorithms. - Promote the ability to analyze and evaluate AI applications in different professional contexts.	- Understand the basic concepts of AI, including machine learning algorithms. - Analyze data using AI tools. - Evaluate the ethical implications of AI in society.	- Understand the basic principles and applications of AI - Gain familiarity with key AI techniques such as machine learning, deep learning, and natural language processing - Develop critical thinking regarding AI's ethical and societal impact	- Understanding the fundamental concepts of AI - Become familiar with basic AI algorithms - Learn to use common AI tools and platforms
Content Description	- Fundamentals of artificial intelligence and its history - Introduction to supervised and unsupervised learning - AI tools and platforms (e.g. Python) - Case studies - Ethical and quality considerations in the use of AI.	Modules: - Introduction to AI (history, types, applications), - Fundamentals of Machine Learning (supervised vs. unsupervised), - Ethics and Regulation in AI	- Introduction to AI and its history - Machine learning basics - Neural networks and deep learning fundamentals - AI applications in industry - Ethical considerations in AI	- Introduction to AI (definitions, history, applications) - Basic Algorithms (supervised learning, unsupervised learning, reinforcement) - AI Tools and Platforms (TensorFlow, PyTorch, etc) - Ethics and AI Challenges
Mode of Instruction	Online, Hybrid, or In-person	Hybrid (20% in-person, 80% virtual)	Online + In-person workshops	Hybrid (online and in-person)
Learning Outcomes	- LO1: Understand the basic concepts of artificial intelligence (Knowledge), ESCO: Analyze information, understand basic algorithms - LO2: Apply basic machine learning techniques to solve simple problems (Capacity), ESCO: Use data analysis tools, design solutions with AI algorithms - LO3: Interpret the results obtained by applying AI algorithms (Capacity), ESCO: Evaluate predictive models, perform critical analysis of results	- Define basic AI concepts (Knowledge), ESCO: AI Fundamentals - Apply simple Machine Learning algorithms (Skill), ESCO: Machine Learning Basics - Evaluate ethical case studies in AI (Competence), ESCO: Ethical AI Practices	- Apply machine learning techniques - Develop AI models - Evaluate AI system performance	- LO1: Identify the different types of AI and their applications (Knowledge), related ESCO skills: Artificial intelligence, Machine learning; - LO2: Implement basic AI algorithms (Ability), related ESCO skills: Algorithm development, Programming; - LO3: Use AI tools to solve simple problems (Ability), related ESCO skills: Data analysis, Problem solving
Experience Level	Beginner	EQF Level 5	Beginner	EQF Level 5
Competency Frameworks	ESCO, EQF	ESCO, DIGCOMP 2.1	ESCO, European AI Competency Framework	ESCO
Learning Activities	- Theoretical classes (webinars, videos) - Practical workshops (virtual laboratories, programming exercises) - Collaborative group projects - Discussion forums and self-assessment activities.	- Theoretical (lectures, webinars with experts) - Practical (virtual labs (Python, TensorFlow), team projects)	- Recorded lectures - Coding exercises - Case studies - Group discussions	- Online lectures - Practical labs - Group projects
Workload (Theoretical)	<Number of theoretical hours>	2 ECTS (50 hours)	15 hours	2 ECTS (50 hours)
Workload (Practical)	<Number of practical hours>	3 ECTS (75 hours)	10 hours	3 ECTS (75 hours)

TABLE IV: Overview of “Assessment and Certification” Definitions

Element	ChatGPT (o3-mini)	DeepSeek-R1	ChatGPT (GPT-4 Turbo)	NotebookLM (Gemini 2.0)
Assessment Methods	- Short exams - Quizzes - Projects - Class participation	- Written exam (30%) - Hands-on project (50%) - Reflective portfolio on AI ethics (20%)	- Online quizzes (40%) - Practical project (40%) - Peer review (20%)	- Written exam (30%) - Practical projects (40%) - Class participation (30%)
List of Trainers	- <Names and positions of trainers>	- <Professor’s name>, PhD in AI (UC3M) - <Professional’s name>, Data Ethics Specialist <Company X>	- <To be determined>	- <Instructor Name>, <Qualification>
Stackability Options	- Cumulative with other micro-credentials - Partial recognition in degrees	- Stack towards Professional Certificate in AI or Master’s in Data Science	- Combine with ML and Data Science for advanced AI certificate	- Combine with other AI micro-credentials for specialization certificate
Outcome or Rights for Student	- It grants recognition of artificial intelligence skills, facilitating academic mobility and employability in technological sectors	- Digital micro-credential via Europass - Career advancement or further education	- Digital credential via Europass	- University Micro-credential - Opportunity for advanced AI training
Quality Assurance	- UC3M Internal Quality Assurance System (SGIC) - European regulations	- Internal Quality Assurance System (SGIC) of UC3M, audited by ANECA - Annual review by the Teaching Innovation Committee	- UC3M internal quality assurance - Compliance with EDC	- UC3M Internal Quality Assurance System (SGIC) - Compliance with ESG
Certifying University	UC3M	Universidad Carlos III de Madrid (UC3M)	Universidad Carlos III de Madrid	Universidad Carlos III de Madrid
Date of Issue	<Micro-credential issue date>	<Date of Issue>	<To be determined>	<Issue Date>

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